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SIMATS ENGINEERING
ASSESSMENT TOOLS 3

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA1301**

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

CLASS TEST

Code of Conduct:

I Deepak V-192411021 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

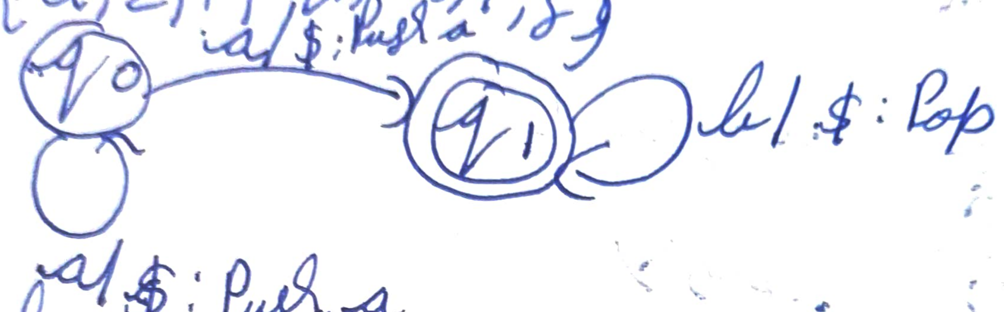
Name : Deepak V
Reg No : 192411021
Course Code : CSA4301

Theory Of Computation CO4 AT3 - Class Test

① DPDA to accept : $L = \{a^n b^n\}\ n \geq 1$

$L = \{a^n b^n\} \Rightarrow \{ab, aabb, aaabbb\}$

$M = \{Q, \Sigma, \Gamma, q_0, z_0, F, \delta\}$



Take aaabbb

no	State	Input symbol	Stack	Action
1	q_0	aaabbb$	$	Start
2	q_0	aabbb $	a$	Push a
3	q_0	abbb $	aa$	Push a
4	q_0	bbb$	aaa$	Push a
5	q_1	bb$	aa$	Pop
6	q_1	b$	a$	Pop
7	q_1	$	$	Pop

$\therefore L = \{a^n b^n\}$ accepted.

2) CFG to PDA for language

Given Language:

$E \rightarrow E+E \mid E*E \mid (E) \mid I$

$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

$M = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0\}$

Transitions:

$\delta(q, \epsilon, E) = (q, E+E)$

$\delta(q, \epsilon, E) = (q, E*E)$

$\delta(q, \epsilon, E) = (q, (E))$

$\delta(q, \epsilon, E) = (q, I)$

$\delta(\cancel{I}\, q, \epsilon, I) = (q, Ia)$

$\delta(q, \epsilon, I) = (q, Ib)$

$\delta(q, \epsilon, I) = (q, I0)$

$\delta(q, \epsilon, I) = (q, I1)$

$\delta(q, \epsilon, I) = (q, a)$

$\delta(q, \epsilon, I) = (q, b)$

$\Gamma = +, *, (,), a, b, 0, 1$

(3)

Matching Terminals:

$\delta(q, +, +) = (q, \epsilon)$

$\delta(q, *, *) = (q, \epsilon)$.

$\delta(q, (, () = (q, \epsilon)$

$\delta(q,),)) = (q, \epsilon)$

$\delta(q, a, a) = (q, \epsilon)$

$\delta(q, b, b) = (q, \epsilon)$

$\delta(q, 0, 0) = (q, \epsilon)$

$\delta(q, 1, 1) = (q, \epsilon)$

$\therefore$ CFG converted to PDA using Transitions

3) TM for $L = \{a^n b^n\}$

1) Take input symbol ~~for~~ from language
2) Start from Left till a, and change to x
3) ~~&~~ after changing, X go from Right till b change to Y if a found reject.

if equal no. accept, else reject.

Eg: aabb

aabb a found

X abb a -> X

X a bb b found

X a b Y b -> Y

X a b Y X found, move Right

X a b Y a found

X X b Y a -> X

X X b Y Y found, move Left

X X b Y b found, ~~mo~~

$\boxed{XXYY}$ b -> Y

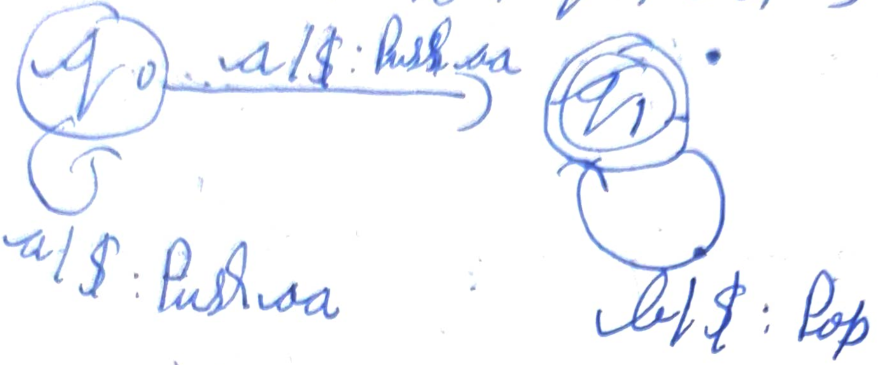
∴ aabb accepted.

i) PDA for $L = \{a^n b^{2n}\}$ $n \geq 1$

$L = \{a^n b^{2n}\} \Rightarrow \{abb, aabbbb, aaabbbbbb \cdots\}$

Take aabbbb

$M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$



no.	State	Input	Stack	Action
1	q_0	aabbbb\$	\$	Start
2	q_0	abbbb\$	aa\$	Push aa
3	q_0	bbbb\$	aaaa\$	Push aa
4	q_1	bbb\$	aaa\$	Pop
5	q_1	bb\$	aa\$	Pop
6	q_1	b\$	a\$	Pop
7	q_1	\$	\$	Pop.

$\therefore$ aabbbb and $L = \{a^n b^{2n} \; n \geq 1\}$ accepted.

5) TM for $L = \{a^n b^n c^n\}$

1) Take input from language

2) Start from left till a, change to X

3) go from right till b, change to Y,
a found reject

4) go from right till c, change to Z, a, b found
reject

5) repeat.

eg. aabbcc , a found, a → X

~~a~~ X abbcc , c found, Move ~~right~~ left

X abbcc , b found, b → Y

X abYcc , c found, c → Z

X abYcZ , X found. Move right

X X bYcZ , a found, a → X

X X YYcZ , b found, b → Y.
c found, c → Z

$\boxed{X X Y Y Z Z}$

aabbcc accepted.

SIMATS ENGINEERING
ASSESSMENT TOOLS

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

88/100

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